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Trackable protective packaging for tools and methods for calibrating tool installation using the same

Abstract

Navigation systems, methods and software for assisting in determining whether a tool is properly installed on a surgical device. A protective packaging retains the tool and has a trackable feature. The trackable feature has a predetermined state defined relative to the tool and the trackable feature is detectable by a localizer. One or more controllers acquire the actual state of the tool based on the detected trackable feature and compare the actual state of the tool with an expected state of the tool, which is based on an expected condition in which the tool is properly mounted to the surgical device. Based on the comparison, the one or more controllers can determine whether the tool is properly mounted to the surgical device.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3495702	12/1969	Kuster	N/A	N/A
4519501	12/1984	Cerwin	N/A	N/A
5220769	12/1992	Brown et al.	N/A	N/A
5246109	12/1992	Markle et al.	N/A	N/A
5392917	12/1994	Alpern et al.	N/A	N/A
5485917	12/1995	Early	N/A	N/A
5542427	12/1995	Aekerfeldt	N/A	N/A
5542539	12/1995	Early	N/A	N/A
5584164	12/1995	Sinn	N/A	N/A
5727682	12/1997	Abidin et al.	N/A	N/A
5727685	12/1997	Aganza et al.	N/A	N/A
5842567	12/1997	Rowe et al.	N/A	N/A
5921992	12/1998	Costales et al.	N/A	N/A
5947288	12/1998	Dykstra et al.	N/A	N/A
5987960	12/1998	Messner et al.	N/A	N/A
6059111	12/1999	Davila et al.	N/A	N/A
6059112	12/1999	Dykstra et al.	N/A	N/A
6161695	12/1999	Nicolais	N/A	N/A
6332891	12/2000	Himes	N/A	N/A
6340363	12/2001	Bolger et al.	N/A	N/A
6412639	12/2001	Hickey	N/A	N/A
6511418	12/2002	Shahidi et al.	N/A	N/A
6517478	12/2002	Khadem	N/A	N/A
6671058	12/2002	Braunecker et al.	N/A	N/A
6814236	12/2003	Roshdy	N/A	N/A
6837027	12/2004	Hickey	N/A	N/A

6866671	12/2004	Tierney et al.	N/A	N/A
6915901	12/2004	Feinberg et al.	N/A	N/A
6996487	12/2005	Jutras et al.	N/A	N/A
7043961	12/2005	Pandey et al.	N/A	N/A
7104401	12/2005	Smith et al.	N/A	N/A
7128208	12/2005	Hull	N/A	N/A
7166114	12/2006	Moctezuma De La Barrera et al.	N/A	N/A
7316318	12/2007	Rosten et al.	N/A	N/A
7320404	12/2007	Landis	N/A	N/A
7331463	12/2007	Hickey	N/A	N/A
7353946	12/2007	Cervantes	N/A	N/A
7451870	12/2007	Donahoe et al.	N/A	N/A
7467710	12/2007	Cerwin et al.	N/A	N/A
7516845	12/2008	Lang et al.	N/A	N/A
7611008	12/2008	Ruffieux et al.	N/A	N/A
7617932	12/2008	Windus-Smith et al.	N/A	N/A
7621395	12/2008	Mogensen et al.	N/A	N/A
7648030	12/2009	Landis	N/A	N/A
7726564	12/2009	Goldbach	N/A	N/A
7770722	12/2009	Donahoe et al.	N/A	N/A
7770728	12/2009	Kærn	N/A	N/A
7820446	12/2009	Feilkas et al.	N/A	N/A
7837694	12/2009	Tethrake	606/79	A61B 90/98
7962192	12/2010	Bodduluri et al.	N/A	N/A
7975842	12/2010	Thoes et al.	N/A	N/A
8006839	12/2010	Hafner	N/A	N/A
8010180	12/2010	Quaid et al.	N/A	N/A
8020703	12/2010	List et al.	N/A	N/A
8042689	12/2010	Frojd et al.	N/A	N/A
D655165	12/2011	Jensen et al.	N/A	N/A
8146329	12/2011	Bryant et al.	N/A	N/A
8172866	12/2011	List	N/A	N/A
8261910	12/2011	Guenter et al.	N/A	N/A
8303599	12/2011	Hess et al.	N/A	N/A
8320612	12/2011	Knobel et al.	N/A	N/A
8365910	12/2012	Valaie et al.	N/A	N/A
8517174	12/2012	Dacey et al.	N/A	N/A
8739801	12/2013	Rains et al.	N/A	N/A
8770405	12/2013	Dacey et al.	N/A	N/A
8838205	12/2013	Shoham et al.	N/A	N/A
8881900	12/2013	Witt et al.	N/A	N/A
8893883	12/2013	Valaie et al.	N/A	N/A
D731326	12/2014	Johansson	N/A	N/A
9068820	12/2014	Kosmecki et al.	N/A	N/A
9089944	12/2014	Schneider	N/A	B24D 13/147
9089952	12/2014	Gatling	N/A	B21J 15/10
9096368	12/2014	Wu	N/A	N/A
9119655	12/2014	Bowling et al.	N/A	N/A

9138296	12/2014	Grabowski	N/A	N/A
9157762	12/2014	Ji	N/A	N/A
9265579	12/2015	Richart	N/A	N/A
9393039	12/2015	Lechner et al.	N/A	N/A
9439658	12/2015	Ford et al.	N/A	N/A
9474686	12/2015	Neal et al.	N/A	N/A
9532847	12/2016	Hendrickson et al.	N/A	N/A
9585727	12/2016	Richart	N/A	N/A
9589226	12/2016	Elizondo, II	N/A	G06K 19/07786
9592098	12/2016	Richart	N/A	N/A
9597092	12/2016	Pernot et al.	N/A	N/A
RE46409	12/2016	Foley et al.	N/A	N/A
9652591	12/2016	Moctezuma de la Barrera et al.	N/A	N/A
9707039	12/2016	Grabowski et al.	N/A	N/A
9717843	12/2016	Grucela et al.	N/A	N/A
9750579	12/2016	Richart	N/A	N/A
9775626	12/2016	Revie et al.	N/A	N/A
9776783	12/2016	Nadig et al.	N/A	N/A
9828157	12/2016	Roesler	N/A	N/A
D806541	12/2017	Love et al.	N/A	N/A
9872754	12/2017	Tuechsen et al.	N/A	N/A
9926116	12/2017	Kinyon	N/A	N/A
D818813	12/2017	Love et al.	N/A	N/A
9975679	12/2017	Hulliger	N/A	N/A
9999469	12/2017	Roesler	N/A	N/A
10004567	12/2017	Dacey et al.	N/A	N/A
10016247	12/2017	Grabowski et al.	N/A	N/A
10017303	12/2017	Wagner et al.	N/A	N/A
10029043	12/2017	Grucela et al.	N/A	N/A
10059499	12/2017	Roesler et al.	N/A	N/A
10086131	12/2017	Okihara	N/A	N/A
10130439	12/2017	Richart	N/A	N/A
10159555	12/2017	Bailly et al.	N/A	N/A
10357184	12/2018	Crawford et al.	N/A	N/A
10456207	12/2018	Flatt	N/A	N/A
10556345	12/2019	Kamikawa et al.	N/A	N/A
2002/0032452	12/2001	Tierney et al.	N/A	N/A
2003/0195526	12/2002	Vilsmeier	N/A	N/A
2005/0187562	12/2004	Grimm et al.	N/A	N/A
2005/0218024	12/2004	Lang et al.	N/A	N/A
2005/0220849	12/2004	Hickey	N/A	N/A
2005/0288575	12/2004	de la Barrera et al.	N/A	N/A
2006/0142657	12/2005	Quaid et al.	N/A	N/A
2006/0200046	12/2005	Windus-Smith et al.	N/A	N/A
2007/0185486	12/2006	Hauck et al.	N/A	N/A
2007/0203393	12/2006	Stefanchik	N/A	N/A
2007/0295620	12/2006	Collet et al.	N/A	N/A

2008/0004633	12/2007	Arata et al.	N/A	N/A
2008/0228195	12/2007	von Jako et al.	N/A	N/A
2009/0163930	12/2008	Aoude et al.	N/A	N/A
2009/0247861	12/2008	Manus et al.	N/A	N/A
2011/0264107	12/2010	Nikou et al.	N/A	N/A
2012/0305427	12/2011	Felder et al.	N/A	N/A
2013/0063254	12/2012	Hanks	235/492	G06K 19/04
2013/0299371	12/2012	Johansson	N/A	N/A
2014/0148808	12/2013	Inkpen et al.	N/A	N/A
2014/0251845	12/2013	Roesler	N/A	N/A
2014/0251846	12/2013	Roesler	N/A	N/A
2015/0182293	12/2014	Yang et al.	N/A	N/A
2015/0297296	12/2014	Stauder et al.	N/A	N/A
2016/0135895	12/2015	Faasse et al.	N/A	N/A
2016/0151119	12/2015	Michel et al.	N/A	N/A
2016/0166350	12/2015	Burkhardt et al.	N/A	N/A
2016/0244234	12/2015	Mayer et al.	N/A	N/A
2016/0249986	12/2015	Kruger et al.	N/A	N/A
2017/0000583	12/2016	Lechner et al.	N/A	N/A
2017/0007388	12/2016	Neal et al.	N/A	N/A
2017/0119487	12/2016	Binder et al.	N/A	N/A
2017/0137194	12/2016	Lorence	N/A	N/A
2017/0143432	12/2016	Bowling et al.	N/A	N/A
2017/0252109	12/2016	Yang et al.	N/A	N/A
2017/0290634	12/2016	Dacey	N/A	N/A
2017/0333052	12/2016	Ding et al.	N/A	N/A
2017/0333138	12/2016	Arata et al.	N/A	N/A
2017/0349347	12/2016	Bentz	N/A	N/A
2017/0355505	12/2016	Nadig et al.	N/A	N/A
2018/0014888	12/2017	Bonny et al.	N/A	N/A
2018/0023946	12/2017	Elliot	N/A	N/A
2018/0140223	12/2017	Kheradpir et al.	N/A	N/A
2018/0162619	12/2017	Kocur et al.	N/A	N/A
2018/0168750	12/2017	Staunton et al.	N/A	N/A
2018/0214167	12/2017	Overmyer et al.	N/A	N/A
2018/0222655	12/2017	Grabowski et al.	N/A	N/A
2018/0235348	12/2017	Booker	N/A	N/A
2018/0296293	12/2017	Jeda	N/A	N/A
2018/0296297	12/2017	Moloney et al.	N/A	N/A
2018/0296749	12/2017	Grucela et al.	N/A	N/A
2018/0311008	12/2017	Grabowski et al.	N/A	N/A
2018/0344422	12/2017	Deck	N/A	N/A
2019/0001050	12/2018	Okihara	N/A	N/A
2019/0070019	12/2018	Frederick et al.	N/A	N/A
2019/0269465	12/2018	Murrell et al.	N/A	N/A
2019/0315563	12/2018	Johansson	N/A	N/A
2020/0038115	12/2019	Schwamb et al.	N/A	N/A
2020/0054199	12/2019	Kambe et al.	N/A	N/A
2020/0121400	12/2019	Girardeau-Montaut et al.	N/A	N/A

2020/0261161	12/2019	Frielinghaus et al.	N/A	N/A
2020/0297429	12/2019	Montane et al.	N/A	N/A
2020/0356077	12/2019	Singer-Schnoeller	N/A	G05B 19/4155

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2298652	12/2010	EP	N/A
2292528	12/2012	EP	N/A
2662312	12/2013	EP	N/A
2011135246	12/2010	WO	N/A
2017087732	12/2016	WO	N/A
2017114059	12/2016	WO	N/A
2017129757	12/2016	WO	N/A
2017157763	12/2016	WO	N/A
2017176569	12/2016	WO	N/A
2017185029	12/2016	WO	N/A
2017185052	12/2016	WO	N/A
2018011257	12/2017	WO	N/A
2018052832	12/2017	WO	N/A
2018106110	12/2017	WO	N/A
2018112107	12/2017	WO	N/A
2018156589	12/2017	WO	N/A
2018156611	12/2017	WO	N/A
2018164497	12/2017	WO	N/A
2018198308	12/2017	WO	N/A
2018202805	12/2017	WO	N/A
2018227066	12/2017	WO	N/A
2019227215	12/2018	WO	N/A

OTHER PUBLICATIONS

English language abstract for EP 2 292 528 extracted from espacenet.com database on Apr. 30, 2020, 1 pages. cited by applicant

English language abstract for WO 2011/135246 extracted from espacenet.com database on Apr. 16, 2020, 2 pages. cited by applicant

English language abstract and machine-assisted English translation for WO 2017/114059 extracted from espacenet.com database on Apr. 16, 2020, 9 pages. cited by applicant

English language abstract and machine-assisted English translation for WO 2018/164497 extracted from espacenet.com database on Apr. 16, 2020, 15 pages. cited by applicant

English language abstract for WO 2018/198308 extracted from espacenet.com database on Apr. 16, 2020, 2 pages. cited by applicant

English language abstract and machine-assisted English translation for WO 2018/202805 extracted from espacenet.com database on Apr. 16, 2020, 14 pages. cited by applicant

International Search Report for Application No. PCT/US2018/027138 dated Jun. 25, 2018, 3 pages. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The subject application is a continuation of U.S. patent application Ser. No. 16/817,118, filed Mar. 12, 2020, which claims priority to and all the benefits of U.S. Provisional Patent App. No. 62/820,577, filed Mar. 19, 2019, the entire contents of each of the aforementioned applications being hereby incorporated by reference.

BACKGROUND

(1) A surgical device, such as a robot, often receives a tool or instrument for use during a surgical procedure. The tool may be a cutting accessory, such as a saw, bur drill, having a head with sharp features configured to resect tissue such as bone. Exposed handling of the tool may result in surgical site infection, injury, and other undesirable consequences. For these reasons, such tools often require a type of protective packaging to prevent inadvertent exposure before and during installation.

(2) A working portion of the tool (e.g., such as the burring portion) must be precisely known by the system in order to effect proper control of the surgical device. Navigation systems are often utilized with surgical systems to track surgical devices, the patient, and additional tools that may be utilized in the surgical procedure. The navigation system often utilizes markers or trackers that are attached to the objects that need to be tracked.

(3) Tool accessories, such as those mentioned above, are interchangeably installed, often operate a high rate of movement and have high exposure to the surgical site thereby requiring sterilization. Therefore, markers or trackers on the tool may negatively affect physical performance of the tool or may be impossible to practically implement on the tool. Furthermore, markers on the tool would likely be destroyed during sterilization. For these reasons, such tool accessories are often not directly tracked by the navigation system.

(4) In such situations, there may be no way of knowing whether the tool is properly installed to the surgical device. There may be situations where the tool appears to be properly installed to the operator, when in fact, the tool may be slightly (e.g., by a few millimeters) mis-aligned or not fully seated.

(5) Prior techniques for confirming tool installation require additional instruments, such as tracked digitization devices, that are used to digitize certain points on the tool for calibration purposes. However, such techniques require additional tools and operator steps thereby increasing operating time and costs. Additionally, digitization techniques may produce measurements that are less accurate because the process requires manual operator involvement in digitizing. For example, the actual digitization point may deviate from the digitization point expected by the system.

(6) Another prior method to assess installation accuracy is to use a custom end effector for a robotic system whereby the end effector has markers or tracking elements detectable by the navigation system and the tool accessory is already in a pre-installed, fixed position, relative to the end effector. In such instances, the robot must move to certain poses to implement a calibration process. In these poses, a kinematic location of the robot is known to the system and the locations of the tracking elements on the end effector are compared to the kinematic robot locations to assess accuracy. However, such methods are not suitable for end effectors that can accept different tools. Furthermore, the end effector must be customized with the given tool and therefore, this increases costs and complexity in end effector versions. Additionally, this technique requires the robot to assume certain poses, thereby increasing operating room time.

(7) Techniques designed to overcome one or more of the aforementioned disadvantages are desired.

SUMMARY

(8) According to a first aspect, a navigation system is provided, comprising: a trackable protective

packaging configured to retain a tool, the trackable protective packaging comprising at least one trackable feature configured to have a predetermined state defined relative to an actual state of the tool when the tool is retained by the trackable protective packaging; a localizer configured to detect an actual state of the at least one trackable feature; and one or more controllers coupled to the localizer and configured to: obtain the predetermined state of the at least one trackable feature; obtain an expected state of the tool based on an expected condition in which the tool is properly mounted to a surgical device; determine the actual state of the tool based on the actual state of the at least one trackable feature detected by the localizer and the predetermined state of the at least one trackable feature defined relative to the actual state of the tool; compare the actual state and the expected state of the tool; and evaluate whether the tool is properly mounted to the surgical device based on comparison of the actual state and expected state of the tool.

(9) According to a second aspect, a method is provided of operating a navigation system, the navigation system comprising a localizer, one or more controllers, and a trackable protective packaging configured to retain a tool and the trackable protective packaging comprising at least one trackable feature, the method comprising: obtaining, with the one or more controllers, a predetermined state of the at least one trackable feature, the predetermined state defined relative to an actual state of the tool when the tool is retained by the trackable protective packaging; obtaining, with the one or more controllers, an expected state of the tool based on an expected condition in which the tool is properly mounted to a surgical device; detecting, with the localizer, an actual state of the at least one trackable feature; determining, with the one or more controllers, the actual state of the tool based on the actual state of the at least one trackable feature detected by the localizer and the predetermined state of the at least one trackable feature defined relative to the actual state of the tool; comparing, with the one or more controllers, the actual state and the expected state of the tool; and evaluating, with the one or more controllers, whether the tool is properly mounted to the surgical device based on comparing the actual state and expected state of the tool.

(10) According to a third aspect, a non-transitory computer-readable medium is provided, comprising instructions, which when executed by one or more processors, are configured to: obtain a predetermined state of at least one trackable feature of a trackable protective packaging that is configured to retain a tool, the predetermined state defined relative to an actual state of the tool when the tool is retained by the trackable protective packaging; obtain an expected state of the tool based on an expected condition in which the tool is properly mounted to a surgical device; acquire an actual state of the at least one trackable feature detected by a localizer; determine the actual state of the tool based on the actual state of the at least one trackable feature and the predetermined state of the at least one trackable feature defined relative to the actual state of the tool; compare the actual state and the expected state of the tool; and evaluate whether the tool is properly mounted to the surgical device based on comparison of the actual state and expected state of the tool.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Advantages of the present disclosure will be readily appreciated as the same becomes better understood by reference to the following detailed description when considered in connection with the accompanying drawings wherein:

(2) FIG. 1 is a perspective view of a protective packaging for retaining a tool in accordance with one example.

(3) FIG. 2 is a plan view of the protective packaging of FIG. 1 further illustrating trackable features on the packaging and their relationship to a tool center point of the tool.

(4) FIG. 3 is an example system comprising a navigation system utilized in conjunction with a surgical device, in this example, the end effector of a robotic manipulator which is configured to

receive the tool retained by the packaging, and further illustrates relationships between components of the system that can be known or determinable by one or more controllers to determine the state of the tool center point of the tool.

(5) FIG. 3A is an expanded view of the packaging and tool from FIG. 3 further illustrating the relationship between the trackable features and the tool center point.

(6) FIG. 4 illustrates one example of surgical device and the one or more controllers wherein an expected tool center point of the tool known to the one or more controllers is illustrated in a hypothetical condition wherein the tool is properly installed to the surgical device.

(7) FIG. 5 illustrates the surgical device of FIG. 4 wherein the packaged tool is improperly installed to the surgical device, and wherein the one or more controllers compares the expected tool center point from FIG. 4 to an actual state of the tool center point to determine an error condition.

DETAILED DESCRIPTION

(8) Described herein are trackable protective packages for tools, and systems and methods for tracking the protective packaging for various purposes, such as to confirm installation of the tool to a surgical device.

(9) A. Examples of Protective Packaging for Tools

(10) Referring to FIG. 1, a non-limiting example of a protective packaging 22 is illustrated. The protective packaging 22 is specifically shaped to accommodate and retain a tool 24 and provides safe, sterile and secure handling of the tool 24 during storage, transport, and mounting of the tool 24 on a surgical device 28 (see FIGS. 3 and 5). In view of the techniques described below, having the protective packaging 22 be utilized during or after mounting of the tool 24 on the surgical device 28 provides significant advantages.

(11) The protective packaging 22 and its uses during tool 24 installation can be like those described in US Pat. App. Pub. No. 2018/0296297A1, entitled “Packaging Systems and Methods for Mounting A Tool on A Surgical Device”, the disclosure of which is hereby incorporated by reference in its entirety. However, the protective packaging 22 can have different configurations from the protective packaging 22 shown.

(12) In one example, the tool 24 is a device that is configured to remove material from a target site, such as a bone of a patient, soft tissue, or the like. For these reasons, the tool 24 is likely to be a sharp object. The tool 24 can also be a passive object, i.e., a purely mechanical object having no actively energizable electrical components. Examples of tools 24 include, but are not limited to burs, drill bits, screw drivers, saws, and the like.

(13) The tool 24 comprises a working portion W or energy applicator. The working portion W is a feature of the tool 24 that is configured to interact with and manipulate the target site. When the tool 24 is a bur, the working portion W is the bur head 26 rigidly coupled to a tool shaft 25 (as shown). When the tool 24 is a drill bit, the working portion W can be the threaded portion of the drill shaft or a distal tip of the drill bit. When the tool 24 is a saw, the working portion W can be a distal tip or teeth of the saw blade. For simplicity in description, the tool 24 described in the examples below is a bur and the working portion is the bur head 26. However, various other tools 24 with different working portions W are fully contemplated to be utilized with the techniques described herein.

(14) The surgical device 28 may be any apparatus configured to receive and operate the tool 24. In other words, the surgical device 28 may provide actuation, control, power, etc., to the tool 24. The surgical device 28 of FIGS. 3-5 is a surgical robot R having an end effector configured to receive the tool 24. In this example, the surgical device 28 can be the robot R and/or the end effector. Other example combinations of the tool 24 and the surgical device 28 are contemplated. For example, possible combinations may include: a saw or a blade configured to be received by a saw driver; a router, a curved bur, or a sleeve connector for a bur configured to be received by a handheld rotary instrument; electrodes configured to be received by a smoke evacuation pencil; a scalpel configured to be received by a scalpel handle; an ultrasonic tip configured to be received by an

ultrasonic aspirator; and an endoscopic shaver or cutter configured to be received by an endo-handpiece. For any of the example above, the surgical device **28** can be a hand-held instrument configured to be supported (against the force of gravity) and manually moveable in space by the hand and arm of a user. It is to be understood that other surgical devices for receiving tools are contemplated. As will be described below, the tool **24** and/or surgical device **28** according to any configuration may be tracked by a navigation system.

(15) In one example, the tool **24** includes a distal region **27** and an attachment portion **29**. The working portion **W** is at the distal region **27** of the tool **24** and the attachment portion **29** is the part of the tool **24** that installs to the surgical device **28**. In some instances, the attachment portion **29** is located a proximal region of the tool **24**, opposite the distal region **27**. At the distal region **27** is a distal end **30** and at the attachment region **29** is a proximal end **32** of the tool **24**. A length of the tool **24** can be defined between the distal end **30** and the proximal end **32**. If the tool **24** is symmetrical about an axis of rotation, a tool axis may be defined between the distal end **30** and the proximal end **32**.

(16) In this example, the protective packaging **22** provides a casing that comprises a distal section **34** that has a cavity to retain the distal region **27** of the tool **24** to protect the operator and prevent contamination. The distal section **34** may comprise distal portions **36**, **38** that collectively retain the tool **24**. The distal portions **36**, **38** can be separate components or can be integrally formed as one component. The distal portions **36**, **38** may be pivotably coupled to one another such that they can open and close, in a clamshell configuration. In such examples, the distal section **34** can be opened to enable removal of the tool **24** from the protective packaging **22** and closed to enable retention of the **24** in the protective packaging **22**. In other examples, the distal portions **36**, **38** can be permanently coupled together, such as by using a high frequency weld, adhesive, integrally formed material, etc. In such examples, the protective packaging **22** can be configured to slide over and off the tool **24**.

(17) In some examples, the protective packaging **22** can have only the distal section **34**. In other examples, such as that shown in FIG. **1**, the protective packaging **22** can optionally include a proximal section **40** that couples to the distal section **34**. The proximal section **40** can be provided with a cavity to retain the proximal end of the tool **24**, such as the tool shaft **25**. The proximal section **40** can be pivotably coupled to the distal section **34** at a hinge **42**. This hinge **42** enables the proximal section **40** to be detached from the tool shaft **25** simultaneously while remaining hinged to the distal section **34**. This way, the tool shaft **25** can be exposed for installation while the proximal section **40** can provide shielding protection for the operator's hand. In other examples, the hinge may comprise a perforation **44** that enables the proximal section **40** to be detached completely from the distal section **34**.

(18) Importantly, the protective packaging **22** provides a low dimensional tolerance and/or tight mechanical fit when retaining the tool **24**. Features may be designed into the tool **24** and/or the protective packaging **22** to achieve the fit. For example, a hole could be placed in a saw blade with a mating protrusion in the protective packaging **22**, or an undercut could of a drill shaft could be used to align with a protrusion in the protective packaging **22**. As will be described below, the tight tolerance securely retains the tool **24** while providing a known relationship between features of the protective packaging **22** (e.g., trackable features) and the tool **24**. In one example, the dimensional tolerance is less than 1 mm or even less than 0.1 mm. Any suitable casing of the protective packaging **22** is contemplated that protect the operator and prevent contamination while maintaining a tight mechanical retention of the tool **24**. Hence, the casing of the protective packing **22** is not limited to the examples described herein.

(19) The protective packaging **22** may comprise, in part or entirely, material such as polyethylene terephthalate glycol-modified (PETG). Other suitable materials may include, without limitation, polymers such as polyethylene terephthalate (PET), high-density polyethylene (HDPE), polyvinyl chloride (PVC), low-density polyethylene (LDPE), polypropylene (PP), and polystyrene (PS),

epoxy and other resins, and malleable metals such as aluminum. The protective packaging 22 can be formed by thermoforming, injection molding, vacuum molding, blow molding, or other manufacturing processes.

(20) Various different configurations of the protective packaging 22 are contemplated. For example, the protective packaging 22 can be hermetically sealed (against liquid and gas). The protective packaging 22 can be reusable (sterilizable) or disposable (single use). The protective packaging 22 can have features, such as texturized features, to enable the operator to easily grip the protective packaging 22 for tool 24 installation or to open and close the distal portions 36, 38. The protective packaging 22 or any of its components can be adjustable in size or dimension, and can be elastic, flexible, or deformable. The protective packaging 22 can be tool 24 specific or generically usable for different tools. As will be described below, the protective packaging 22 can also have features that enable the protective packaging 22 to be detectable and/or to avoid being detectable.

(21) The protective packaging 22 can be used for initial attachment of the tool 24 to the surgical device 28 or it can be used for re-attachment of the tool 24 during the surgical procedure. There may be a need to re-check the position of the tool 24 during the procedure or the tool 24 may be removed for a portion of the procedure and later re-installed. The protective packaging 22 can have any combination of the features described herein and can have configurations or features other than those described herein.

(22) B. Trackable Features for the Protective Packaging

(23) As shown in the non-limiting example of FIGS. 1 and 2, the protective packaging 22 can comprise one or more trackable features T. The trackable features T can be utilized to track the tool 24 when the tool 24 is retained by the protective packaging 22. The trackable features T can also be utilized to track the protective packaging 22 when the tool 24 is not retained therein. In another example, described below, the trackable features T are utilized to determine whether or not the tool 24 is properly mounted/installed to the surgical device 28.

(24) The protective packaging 22 may comprise any number and any configuration of trackable features T. In this example, three trackable features T1-T3 are provided with the protective packaging 22. Having at least three trackable features T provides the ability to know the position and orientation of the protective packaging 22. The trackable features T can be placed on or affixed to the protective packaging 22 utilizing any suitable technique. Alternatively, the trackable features T can be integrally formed on, in, or with the protective packaging 22. For example, the trackable features T can be embedded within the material of the protective packaging 22. The trackable features T can also be uniquely identifiable features integral to the protective packaging 22, such as edges, protrusions, recesses, shapes or any of the components of the protective packaging 22 described herein.

(25) The trackable features T can be passive or actively energized and can be of any appropriate type of detectable configuration. The trackable features T may be infrared trackable features. When infrared, the trackable features T can be retroreflective elements or LED emitters, for example. Alternatively or additionally, the trackable features T can be passive landmarks, patterns, and/or shapes that are uniquely identifiable. The trackable features T can also be configured with patterns, colors, gradients, and/or texturized materials. In other examples, the trackable features T can be passive or active magnetic or electro-magnetic elements, passive or active radio frequency elements, radio-opaque elements, ultrasound detectable elements, or any combination of the above-described configurations.

(26) The trackable features T can be provided on the protective packaging 22 in any location, manner, or configuration. For example, the trackable features T can be positioned the first and/or second distal portions 36, 38. The trackable features T can also be positioned on the proximal section 40. In one example, the trackable features T are provided only on the distal section 34, but not the proximal section 40. This configuration is based on the practical consideration that the

working portion W of the tool **24** will remain retained by the distal section **34** during tool **24** installation, whereas the proximal section **40** may be detached from the protective packaging **22** or otherwise pivoted relative to the distal section **34** prior to installation. Furthermore, the trackable features T can be positioned on one side or multiple sides of the protective packaging **22**, including any of the top, bottom, or left or right sides, as needed, to enable the trackable features T to be detectable.

(27) In another example, each of the first and second distal portions **36**, **38** comprises an exterior surface, which is exposed when the protective packaging **22** is closed. In this example, one or more trackable features T can be coupled to the exterior surface of at least one of the first and second distal portions **36**, **38**. As such, the trackable features T can be readily visible when the protective packaging **22** is closed.

(28) In yet another example, the trackable features T may comprise sub-components on different parts of the protective packaging **22**. For example, each of the first and second distal portions **36**, **38** comprises an interior surface. The interior surfaces engage each other when the protective packaging **22** is closed. In this example, one or more trackable features T comprise a first component coupled to the interior surface of the first distal portion **36** and a second component coupled to the interior surface of the second distal portion **38**. The first and second components are positioned such that when the protective packaging **22** is closed, the first and second components engage or combine to form the trackable feature T. In other words, the trackable feature T may be functional when the protective packaging **22** is fully closed and non-functional when the protective packaging **22** is partially or entirely open. The first and second components can be electrical, magnetic, or layers of material (e.g., reflective material) that work together to form the trackable feature T when the protective packaging **22** is closed. The trackable features T can also extend through the material of the protective packaging **22** such that the trackable feature T is on both the exterior and interior surfaces of the protective packaging **22**.

(29) The protective packaging **22** may comprise a material that is transparent to light. The transparent material can increase visibility and/or reduce retro-reflectivity of the protective packaging **22** to increase accuracy in detecting the trackable features T. When transparent, visible light penetration through the protective packaging **22** material may be greater than 75%, 90%, or 95%. The protective packaging **22** can also be equipped with surface properties or features to reduce retro-reflectivity. Examples of such properties or features include, but are not limited to, rough surfaces, dimpled surfaces, anti-reflective coating or film, or the like.

(30) The trackable features T and the configuration of the protective packaging **22** can be different than the examples described above.

(31) C. Example Navigation Systems

(32) To implement tracking of the trackable features T, a navigation system **100** is provided, as shown in FIG. **3**, for example. The navigation system **100** can be utilized in conjunction with the surgical device **28**. As shown in FIG. **3**, for example, the surgical device **28** is the end effector of the robot R that is configured to receive the tool **24**.

(33) The navigation system **100** is configured to track movement of various objects. Such objects include, for example, the protective packaging **22**, the robot R, the surgical device **28** and the anatomy of a patient. The navigation system **100** tracks these objects with respect to a (navigation) localizer coordinate system LCLZ. Coordinates in the localizer coordinate system LCLZ may be transformed to the robot (manipulator) coordinate system MNPL, and/or vice-versa, using transformation techniques described herein.

(34) One example of the navigation system **100**, surgical robot, control techniques, and transformations associated therewith, which can be utilized with the techniques herein, is described in U.S. Pat. App. Pub. No. 2018/0168750A1 filed on Dec. 13, 2017, entitled, "Techniques for modifying tool operation in a surgical robotic system based on comparing actual and commanded states of the tool relative to a surgical site," the entire disclosure of which is hereby incorporated by

reference.

(35) The navigation system **100** may include a cart assembly **102** that houses the one or more controllers **104**, such as a navigation computer, and/or other types of control units. The one or more controllers **104** may also be located in the surgical device **28** and/or a cart of the robot R. The one or more controllers **104** may be located in components or subsystems other than that shown in FIG. **3** and may be implemented on any suitable device or devices in the system **100** other than the configuration shown. In one example, the controllers for the navigation system and the robot are two different controllers performing different operations, and can be configured, for example, as described in U.S. Pat. App. Pub. No. 2018/0168750A1. The one or more controllers **104** can comprise software and/or hardware configured to perform all the tasks described herein. Example of hardware may comprise processors, CPUs, microprocessors, integrated circuits, non-transitory memory, graphic processing units, hard drives, and input/output devices.

(36) A navigation interface is in operative communication with the one or more controllers **104**. The navigation interface includes one or more displays **106**. The one or more controllers **104** is capable of displaying graphical (actual or virtual) representations of the relative states of the tracked objects to the operator using the one or more displays **106**. As will be described below, alerts, notifications, or error messages can also be represented on one or more of the displays **106**.

(37) In addition to the trackable features T described with respect to the protective packaging **22**, the other objects tracked by the navigation system **100** include one or more trackers or trackable features. In one example, as shown, the trackers may include a tool tracker **108** attached to surgical device **28** (e.g., end effector), a robot tracker **109** attached to the base of the robot R (as shown) or to one or more links of the robot R, and target site trackers (not shown), which can be attached to a patient. Any one or more of the trackers may include active or passive trackable features, and may be used with any tracking modality described herein.

(38) The navigation system **100** also includes a navigation localizer **110** (hereinafter “localizer”) that tracks a state of trackable features T on the protective packaging **22**, and other trackers or trackable features. As used herein, the state of an object includes, but is not limited to, data that defines the position and/or orientation of the tracked object or equivalents/derivatives of the position and/or orientation. For example, the state may be a pose of the object, and may include linear data, and/or angular velocity data, and the like. The localizer **110** provides the state of these tracked objects to the one or more controllers **104** to enable the one or more controllers **104** to make determinations based on such information.

(39) In one example, as shown in FIG. **3**, the localizer **110** is an optical localizer and includes a camera unit **112** with one or more optical sensors **114**. The localizer **110** can be an infrared based localizer suitable for detecting trackable features T that are configured with active or passive infrared elements. Although one embodiment of the navigation system **100** is shown in FIG. **3**, the navigation system **100** may have any other suitable configuration for tracking the trackable features T.

(40) Additionally, or alternatively, the localizer **110** can be a machine vision system. The machine vision system can comprise a camera that is configured to detect trackable features T that can include patterns, shapes, colors, textures, gradients and/or uniquely identifiable features provided by the protective packaging **22**. The machine vision system can employ various types of imaging and imaging processing modalities, such as 2D/3D visible light imaging, depth maps, pixel analysis, edge detection, pattern recognition, and the like. One example of a machine vision system that can be utilized is described in US Publication No. 2017/0143432A1, entitled, “Systems and Methods for Establishing Virtual Constraint Boundaries,” the entire contents of which are incorporated by reference herein.

(41) In another example, the navigation system **100** and/or localizer **110** are radio frequency (RF)-based. For example, the navigation system **100** may comprise an RF transceiver in communication with the one or more controllers **104**. The trackable features T of the protective packaging **22** may

comprise RF emitters or transponders. The RF emitters or transponders may be passive or actively energized. The RF transceiver transmits an RF tracking signal and generates state signals to the one or more controllers **104** based on RF signals received from the RF emitters. The one or more controllers **104** may analyze the received RF signals to associate relative states thereto. The RF signals may be of any suitable frequency. The RF transceiver may be positioned at any suitable location to effectively track the objects using RF signals. Furthermore, the RF emitters or transponders may have any suitable structural configuration that may be much different than the trackers or trackable features as shown in the Figures.

(42) The navigation system **100** and/or localizer **110** can also be electromagnetically based. For example, the navigation system **100** may comprise an EM transceiver coupled to the one or more controllers **104**. The trackable features T of the protective packaging **22** may comprise EM components, such as any suitable magnetic tracker, electro-magnetic tracker, inductive tracker, or the like. The trackable features T may be passive or actively energized. The EM transceiver generates an EM field and generates state signals to the one or more controllers **104** based upon EM signals received from the trackers. The one or more controllers **104** may analyze the received EM signals to associate relative states thereto. Again, such navigation system **100** embodiments may have structural configurations that are different than the navigation system **100** configuration as shown throughout the Figures.

(43) In another example, the navigation system **100** and/or localizer **110** are ultrasound-based. For example, the navigation system **100** may comprise an ultrasound imaging device coupled to the one or more controllers **104**. The ultrasound imaging device images any of the aforementioned objects, e.g., the protective packaging **22** and/or trackable features T generate state signals to the one or more controllers **104** based on the ultrasound images. For example, the ultrasound imaging device may be a portable device whose position is tracked. The portable device can be positioned proximate to any object to track the object. The one or more controllers **104** may process the images in near real-time to determine states of the objects. The ultrasound imaging device may have any suitable configuration and may be different than the camera unit **112** as shown in FIG. 3.

(44) The navigation system **100** and/or localizer **110** may be based on any combination of the tracking modalities above and may have any other suitable components or structure not specifically recited herein. Furthermore, any of the techniques, methods, and/or components described above with respect to the camera-based navigation system **100** shown throughout the Figures may be implemented or provided for any of the other embodiments of the navigation system **100** described herein.

(45) D. Tool Center Point and Relationship to Protective Packaging

(46) According to the techniques described herein, the trackable features T of the protective packaging **22** are utilized to track the tool **24** when the tool **24** is retained by the protective packaging **22**. More specifically, the navigation system **100** can track the trackable features T to determine whether or not the tool **24** is properly mounted/attached to the surgical device **28**.

(47) To help facilitate this technique, the protective packaging **22** provides a low dimensional tolerance and tight mechanical fit when retaining the tool **24**. Furthermore, the protective packaging **22** is involved with the process of installing the tool **24** to the surgical device **28**. Therefore, tracking the protective packaging **22** to determine whether or not the tool **24** is properly mounted/attached to the surgical device **28** provides the advantages of reducing steps and reducing additional devices in the operating room. Furthermore, the tight tolerance provides measurements with a high degree of accuracy as compared with prior techniques. The tight tolerance provided by the protective packaging **22** enables the techniques herein to treat the protective packaging **22** as a virtual extension of the tool **24** geometry for calibration or verification purposes. By providing tracking features on the protective packaging **22**, permanent tracking features on the tool **24** (particularly near the working portion) can be avoided. Avoiding permanent tracking features on the tool **24** is advantageous since permanent tracking features on a tool **24** can interfere with tool

24 operation or with the surgical site and cannot be located proximate enough to working portion of the tool **24** since the working portion is utilized to manipulate the anatomy. On the other hand, the protective packaging **22** is installed on the working portion, and hence, maximizes the calibration and verification measurement capability to the distal-most portion of the tool **24**. Furthermore, the methods described herein provide a seamless user experience, as accuracy checks can be done without adding external components or steps.

(48) Referring to FIG. 2, the tool **24** comprises a tool center point (TCP), which in one embodiment, is a predetermined reference point defined at the working portion W or relative to the working portion W. In one embodiment, the TCP is assumed to be located at the center of a spherical of the tool **24** such that only one point is tracked. The TCP may relate to a bur having a specified diameter. Often, the TCP is defined at a location that is at a center of the working portion W. For example, as shown in FIG. 2, the TCP is located at the spherical center of the bur. However, this may not always be the case. For example, the TCP of a drill bit can be a point in the cylindrical center of the drill, but located anywhere along the tool shaft **25**. Similarly, for a saw blade, the TCP can be located anywhere on the saw blade. The TCP can also be arbitrarily defined somewhere with respect to the working portion W. Therefore, the term “center” is not limited to the geometrical center of the working portion W. The TCP may be defined according to various manners depending on the configuration of the working portion W.

(49) Furthermore, the TCP can be a physical point on the working portion W or the TCP can be a virtual point. In either instance, the one or more controllers **104** is configured to store the predetermined TCP state relative to known geometry for the specified tool **24**. Aside from the techniques described herein for confirming installation of the tool **24**, the TCP can also be utilized by the one or more controllers **104** to facilitate control of the tool **24** and/or surgical device **28**. For example, such control can be movement of the tool **24** relative to virtual boundaries or tool paths. Examples of TCPs and uses thereof in controlling surgical devices can be like those described in US Pat. App. Pub. No. 2018/0168750A1, entitled “Techniques for modifying tool operation in a surgical robotic system based on comparing actual and commanded states of the tool relative to a surgical site,” the entire contents of which are hereby incorporated by reference.

(50) In instances where the tool **24** is not directly tracked by the navigation system **100**, the state of the working portion W relative to the surgical device **28** or relative to the navigation system **100** may not be known. Advantageously, when the tool **24** is retained by the protective packaging **22**, the actual state of the TCP can be determined since the protective packaging **22** has trackable features T. As used herein, the “actual” state of the TCP is the real position and/or orientation of the TCP in physical 3D space or in a global coordinate system. In one example, this global coordinate system is the localizer coordinate system LCLZ. To facilitate this determination, the trackable features T are specifically positioned on the protective packaging **22** with a predetermined state defined relative to the actual state of the TCP. The one or more controllers **104** is configured to store these predetermined states of the trackable features T relative to the actual state of the TCP for further evaluations, as will be described below.

(51) Prior to tracking, the actual state of the TCP can be assumed in the correct state relative to the trackable features T based on the understanding that the tool **24** is properly retained by the protective packaging **22**. Mainly, the various features of the protective packaging **22** described above (e.g., tight mechanical fit) help to ensure that the tool **24** is properly retained relative to the protective packaging **22** so as to validate this assumption.

(52) As shown in the example in FIG. 2, the trackable features T1-T3 (in this case, passive markers) and the actual state of the TCP can be defined in a common coordinate system (e.g., the coordinate system of the protective packaging **22**). In this example, the actual state of the TCP can be considered the origin of the coordinate system, and the center of each trackable feature T is utilized for the coordinate measurements. However, it is not necessary to define the actual state of the TCP at the origin of the coordinate system nor to utilize the center of each trackable feature T

for measurement.

(53) In FIG. 2, an X-axis and Y-axis define the coordinate system, where X and Y are distances relative to the origin TCP. There may also be Z-axis coordinates (in and out of the page of FIG. 2) that can be considered in this coordinate system. Z-axis coordinates would be appropriate if any of the trackable features T are defined on a different plane relative to the actual state of the TCP or relative to each other. For simplicity in description and illustration, the Z-axis coordinates are omitted on the assumption that the trackable features T are coplanar with the actual state of the TCP. However, the manner in which the Z-axis coordinates are determined can be like that described herein relative to the X and Y-axis coordinates.

(54) Here, trackable feature T1 is defined at $-X1, Y1$ relative to the actual state of the TCP, trackable feature T2 is defined at $-X1, -Y3$ relative to the actual state of the TCP, and trackable feature T3 is defined at $X1, Y2$, relative to the actual state of the TCP. The coordinate measurements and trackable feature T placement relative to the actual state of the TCP in FIG. 2 is only one example. Of course, the trackable features T can have other predetermined states relative to the actual state of the TCP depending on factors such as the nature of the TCP, the types and positions of the trackable features T, and the structure of the protective packaging 22, etc.

(55) E. Techniques for Evaluating Tool Installation with Tracked Packaging

(56) Referring back to FIG. 3, the navigation system 100 is shown with the tool 24 coupled to the surgical device 28. The operator installs the tool 24 to the surgical device 28 by holding the protective packaging 22. In this example, it is assumed that an attempt was made to install the tool 24 to the surgical device 28, however, the operator may not be aware whether the tool 24 is properly installed.

(57) As can be understood from the example of FIG. 2, the state of the actual TCP is known relative to the states of the trackable features T. As will be described below, the trackable features T of the protective packaging 22 and this known relationship are utilized by the navigation system 100 and one or more controllers 104 for determining whether the tool 24 is properly mounted/installed to the surgical device 28.

(58) To facilitate this determination, the navigation system 100 is shown with various transforms (T) for determining the actual state of the TCP in a global coordinate system, e.g. LCLZ. The transform (T), when calculated, gives the state (position and/or orientation) of a first component in its respective coordinate system relative to the state of a second component in its respective coordinate system. The one or more controllers 104 calculates/obtains and combines a plurality of transforms (T) from the various components of the system, e.g., for purposes such as to validate installation of the tool 24 and/or to control the surgical device 28 or robot R when the tool 24 is installed. The transforms (T) are computational determinations but are represented in FIG. 3 by arrows between the subject components for illustrative purposes. The directionality of the arrow head is not intended to limit the direction of the transform. In other words, the transform (T) can give the state of one component with respect to the other, or vice versa.

(59) In one embodiment, each transform (T) is specified as a transformation matrix, such as a 4×4 homogenous transformation matrix. The transformation matrix, for example, includes three unit vectors representing orientation, specifying the axes (X, Y, Z) from a first coordinate system expressed in coordinates of a second coordinate system (forming a rotation matrix), and one vector (position vector) representing position using the origin from the first coordinate system expressed in coordinates of the second coordinate system.

(60) Example systems and methods for obtaining and computing transforms of the various components of the system is explained in U.S. Pat. No. 9,119,655, entitled, "Surgical Manipulator Capable of Controlling a Surgical Instrument in Multiple Modes," and U.S. Pat. App. Pub. No. 2018/0168750A1 filed on Dec. 13, 2017, entitled, "Techniques for modifying tool operation in a surgical robotic system based on comparing actual and commanded states of the tool relative to a surgical site," the entire disclosures of which are hereby incorporated by reference.

(61) As shown in FIG. 3, the transforms include a first transform (T1) between the localizer **110** and trackable features T of the protective packaging **22** and a second transform (T2) (best shown in FIG. 3A) between the trackable features T and the actual state of the TCP. Other transforms that may be utilized include a third transform (T3) between the localizer **110** and the surgical device **28** tracker **108**, a fourth transform (T4) between the surgical device **28** tracker **108** and a reference point on the surgical device **28**, a fifth transform (T5) between the actual state of the TCP and the reference point on the surgical device **28**, and/or a sixth transform (T6) between the reference point on the surgical device **28** and a base of the robot (R).

(62) Transforms (T1) and (T3) are obtained using tracking data from the localizer **110**. In other words, the localizer **110** tracks the state of the trackable features T of the protective packaging **22** and the state of the surgical device **28** tracker **108** relative to the LCLZ coordinate system. Transform (T2) is obtained based on the known relationship between the trackable features T of the protective packaging **22** and the actual state of the TCP of the tool **24** retained by the protective packaging **22**, as described in the prior section. Transform (T4) is obtained based on a known relationship between the surgical device **28** tracker **108** and the reference point on the surgical device **28**. Transform (T5) is obtained based on a known relationship between an expected state of the TCP (i.e., the state of the TCP when the tool **24** is installed properly, which will be described further below) and the reference point on the surgical device **28**. Transform (T6) can be determined based on kinematic data of the robot (R) stored by the one or more controllers **104** without intervention from the navigation system **100**. Alternatively or additionally, transform (T7) can be obtained by localizer **110** determining the relationship between the surgical device **28** tracker **108** and the tracker **109** at the base of the robot R. Not all of these transforms (T) may be utilized in every instance, and these transforms (T) can be utilized in various combinations.

(63) Any known relationship data (e.g., for transforms T2, T4 or T5) is stored by the one or more controllers **104** and may be fixed (constant or static) or variable. In embodiments where the known relationship data is fixed, the known relationship data may be derived from calibration information relating to the respective components (e.g., the surgical device **28**, the surgical device **28** tracker **108**, and the expected state of the TCP). For example, the calibration information may be obtained at a manufacturing/assembly stage, e.g., using coordinate measuring machine (CMM) measurements, etc. The known relationship data may be obtained using any suitable method, such as reading the known relationship data from a computer-readable medium, an RFID tag, a barcode scanner, or the like. The known relationship data may be imported into system **100** at any suitable moment such that the known relationship data is readily accessible by the one or more controllers **104**. In embodiments where the known relationship data is variable, the known relationship data may be measured or computed using any ancillary measurement system or components, such as additional sensors, trackers, encoders, or the like. The known relationship data may also be acquired after mounting the surgical device **28** tracker **108** to the surgical device **28** in preparation for a procedure by using any suitable technique or calibration method.

(64) With reference to FIGS. 4 and 5, the one or more controllers **104** is configured to evaluate the information described above to determine whether the tool **24** is properly mounted to the surgical device **28**. For example, the one or more controllers **104** can determine whether or not the TCP is in the proper position and take action in response.

(65) In FIG. 4, the one or more controllers **104** is configured to store the expected state of the TCP based on an expected condition in which the tool **24'** is properly mounted to the surgical device **28**. The expected state of the TCP is a virtual state (e.g., point) stored in one or more controllers **104** memory rather than a physical state on the tool **24** itself. For this reason, a dotted representation of the tool **24'** is shown in FIG. 4 to represent the expected state of the TCP during an assumed proper installation. Furthermore, in FIG. 4, a proper installation line **120** is shown within the tool receiving portion of the surgical device **28** to illustrate one example of where a proximal end **32** of the tool **24** should be relative to the surgical device **28** when the tool **24'** is properly installed. In this

illustration, the hypothetical tool **24'** is fully seated within the surgical device **28** (and hence properly installed) such that the proximal end **32** of the tool **24'** aligns with line **120**.

(66) This installation representation is for simplicity in description and illustration, and tool **24'** installation schemes will vary depending on the type of tool being installed and the configuration of the surgical device **28**. Hence, the use of an “installation line” **120** is not intended to limit the nature of proper installation conditions. Indeed, the proper installation between the tool **24** and the surgical device **28** may be defined by any suitable number of reference datum having any complexity designated by the respective installation scheme. Additionally, or alternatively, proper installation may be defined by any suitable physical parameter measured between the tool **24** and the surgical device **28**, wherein the physical parameter can be correlated with, compared to, or otherwise associated to physical parameters derivable from the trackable features of the packaging **22** and the TCP. Such physical parameters may include, but not limited to: position, magnitude of displacement, distance, orientation, linear motion or velocity, angular motion or velocity, inertial parameters, force parameters, displacement parameters, pressure parameters, torque parameters. Hence, instead of comparing the expected state of the TCP to the actual state of TCP based on static position, the techniques may be utilized to compare an expected motion of the TCP during expected proper installation to actual motion of the TCP during actual installation. Accordingly, the term “state” is not intended to be limited to a static moment in time, but may encompass a period of time. Hence, the term “state” can be static or dynamic.

(67) Furthermore, the installation scheme between the tool **24** and the surgical device **28** may include any suitable sensing devices, including but not limited to, direct (DC) electrical connection sensors, proximity sensors, hall effect sensors, electromagnetic sensors, radio frequency sensors, inductive sensors, capacitive sensors, and the like. These sensors are not intended to replace the techniques described herein, but rather can optionally be utilized in conjunction with the techniques described herein, for purposes such as tool identification, coarse tool installation, or tool presence detection.

(68) The expected state of the TCP can be locally derived from the described known relationship data for transform **T5**, which can be obtained using any manner described above. Additionally, or alternatively, the expected state of the TCP can be based on separate known relationship data obtained about the relationship between the expected state of the TCP and any other portion of the surgical device **28**, such as the portion that receives the tool **24**. For any given pose of the robot **R** or state of the surgical device **28**, the one or more controllers **104** can determine the expected state of the TCP based on combining the transforms (**T**) described above. For example, the expected state of the TCP can be determined by combining any one of the following combinations: (**T3+T4+T5**), (**T5+T6**), (**T5+T6+T7**), etc.

(69) If/when the tool **24** is properly installed, the expected TCP and the actual TCP states should be identical. Otherwise, if the tool **24** is improperly installed, the expected TCP and the actual TCP states will be different. Additionally, the trackable features may be different for different tools **24**. This will allow the system to further identify the tool **24** being used and/or confirm the correct tool is used.

(70) FIG. 5 illustrates the actual installation condition of the tool **24** relative to the surgical device **28**. In this situation, the tool **24** is improperly mounted to the surgical device. This error condition is illustrated by distance **d** between the proximal end **32** of the tool **24** and the proper installation line **120**. In other words, the tool **24** is not fully seated in the surgical device **28**. The distance **d** can be difficult to detect to the human eye, e.g., 1 mm or less. Hence, the tool **24** is not properly installed, but may appear so to the operator. Although the error condition is represented by distance **d** in this example, it is possible that the error condition can take other forms depending on the nature of the tool **24**, the surgical device **28**, and/or manner of installation. For example, the error condition can be an identification of an incorrect tool, improper rotation, linear position, orientation, damage to the tool, or any combination thereof. Regardless of the nature of the error

condition, the techniques described herein are configured to detect the error.

(71) Since the tool **24** is improperly installed in FIG. 5, the state of the actual TCP differs from the state of the expected TCP. In this example, the error condition (distance d) at the proximal end **32** of the tool **24** causes a corresponding gap d between the actual TCP and expected TCP states. Of course, other error conditions are possible. For example, a measured distance of the tool **24** that is offset with the tool insertion axis could indicate the tool **24** is damaged or is the incorrect tool.

(72) The one or more controllers **104** can make this determination by obtaining the state of the actual TCP using the transforms (T) described above. Regardless of the pose of the robot R or the state of the surgical device **28**, the one or more controllers **104** can determine the actual TCP state based on combining transformations ($T1+T2$). In other words, the actual state of the TCP is known to the one or more controllers **104** by the tracked state of the trackable features T of the protective packaging **22** and the predetermined states of the trackable features (T) defined relative to the actual state of the TCP. The one or more controllers **104** can arrive at the actual and expected TCP states by combining other combinations of transformations other than those specifically described.

(73) Knowing the states of the actual and expected TCPs, the one or more controllers **104** can compare them using any suitable method, such as comparing position/orientation matrix data from the transforms, etc. In certain examples, different transformation combinations can be compared to validate comparison results. For example, the state of the actual TCP can be validated in the LCLZ coordinate system by comparing the tracking data of the surgical device **28** tracker **108** (transform $T3$) with the tracking data of the trackable features T on the protective packaging **22** (transform $T1$).

(74) If the one or more controllers **104** determines that the states of the actual and expected TCPs, are different, the one or more controllers **104** may instruct generation of a notification or alert to inform the operator of the error condition. The alert or notification can be visual and provided on the display **106**, as shown in FIG. 3. The alert can also be audible or haptic. Based on the nature of the error condition determined by the one or more controllers **104**, the one or more controllers **104** can issue specific instructions about how to resolve the error condition given the circumstance. For example, the one or more controllers **104** may generate a notification instructing the operator to “fully insert the tool shaft” or “rotate the tool shaft 180 degrees clockwise”. At any point, the one or more controllers **104** can re-assess the states of the actual and expected TCPs and inform the operator that the tool **24** is properly installed if the results of the comparison are favorable.

(75) If the expected TCP state is different from the actual TCP state, the system may propose a decision process. For example the system may ask the user to confirm the correct tool is used, the system may suggest the user replace a damaged tool, and/or the user could instruct the system to accept the position of the actual state of the TCP and use the location for the remainder of the surgical procedure.

(76) In certain instances, the one or more controllers **104** can be configured with a tolerance or threshold or range of error or acceptance when evaluating the differences between the states of the actual and expected TCPs. For instance, the one or more controllers **104** can have a tolerance of $+/-0.1$ mm such that if the states of the actual and expected TCPs are off by less than 0.1 mm, the one or more controllers **104** can nevertheless determine that the tool **24** is properly installed. The tolerance may be different from the value given, depending on many factors, such as the mechanical stack-up of components of the system, the nature of the installation, the accuracy of the localizer **110**, or the like. If the actual state of the TCP falls within acceptable range of the expected state of the TCP tolerance, the state of the actual TCP can be stored by the one or more controllers **104** as the calibrated TCP state that will be used for controlling the surgical device **28**.

(77) Several embodiments have been discussed in the foregoing description. However, the embodiments discussed herein are not intended to be exhaustive or limit the invention to any particular form. The terminology which has been used is intended to be in the nature of words of

description rather than of limitation. Many modifications and variations are possible in light of the above teachings and the invention may be practiced otherwise than as specifically described.

Claims

1. A navigation system comprising: a trackable protective packaging configured to retain a tool, the trackable protective packaging comprising at least one trackable feature configured to have a predetermined state defined relative to an actual state of the tool when the tool is retained by the trackable protective packaging; a localizer configured to detect an actual state of the at least one trackable feature; and one or more controllers coupled to the localizer and configured to: obtain the predetermined state of the at least one trackable feature; obtain an expected state of the tool based on an expected condition in which the tool is properly mounted to a surgical device; determine the actual state of the tool based on the actual state of the at least one trackable feature detected by the localizer and the predetermined state of the at least one trackable feature defined relative to the actual state of the tool; compare the actual state and the expected state of the tool; and evaluate whether the tool is properly mounted to the surgical device based on comparison of the actual state and expected state of the tool, wherein the one or more controllers are further configured to generate at least one of a haptic, audible, and visual alert in response to evaluation of whether the tool is properly mounted to the surgical device.
2. The navigation system of claim 1, wherein: the at least one trackable feature comprises at least one passive optical tracking element or at least one active optical tracking element; and the localizer comprises an optical sensor.
3. The navigation system of claim 1, wherein: the at least one trackable feature comprises one or more of a predetermined pattern, a shape and a color; and the localizer is a machine vision system.
4. The navigation system of claim 1, wherein the at least one trackable feature is at least one radio-frequency tracking element or at least one electromagnetic tracking element.
5. The navigation system of claim 1, wherein the trackable protective packaging is configured to retain and substantially enclose a working end of the tool to prevent exposure of the working end, and wherein the trackable protective packaging is configured to be grasped by a user to mount the tool to the surgical device.
6. The navigation system of claim 1, wherein the trackable protective packaging comprises a distal section configured to retain a working portion of the tool in a fixed position relative the distal section, and wherein the distal section includes the at least one trackable feature.
7. The navigation system of claim 6, wherein the trackable protective packaging further comprises a proximal section coupled to the distal section, and the proximal section configured to receive an attachment portion of the tool, and wherein the proximal section is detachable from the distal section by a perforation.
8. The navigation system of claim 6, wherein the at least one trackable feature is provided exclusively on the distal section.
9. The navigation system of claim 1, wherein: the localizer is further configured to track a state of the surgical device; and the one or more controllers obtain the expected state of the tool by further being configured to combine the tracked state of the surgical device and known relationship data between the expected state of the tool and a predetermined reference point on the surgical device.
10. A method of operating a navigation system, the navigation system comprising a localizer, one or more controllers, and a trackable protective packaging configured to retain a tool and the trackable protective packaging comprising at least one trackable feature, the method comprising: obtaining, with the one or more controllers, a predetermined state of the at least one trackable feature, the predetermined state defined relative to an actual state of the tool when the tool is retained by the trackable protective packaging; obtaining, with the one or more controllers, an expected state of the tool based on an expected condition in which the tool is properly mounted to a

surgical device; detecting, with the localizer, an actual state of the at least one trackable feature; determining, with the one or more controllers, the actual state of the tool based on the actual state of the at least one trackable feature detected by the localizer and the predetermined state of the at least one trackable feature defined relative to the actual state of the tool; comparing, with the one or more controllers, the actual state and the expected state of the tool; evaluating, with the one or more controllers, whether the tool is properly mounted to the surgical device based on comparing the actual state and expected state of the tool; and generating, with the one or more controllers, at least one of a haptic, audible, and visual alert in response to evaluating whether the tool is properly mounted to the surgical device.

11. The method of claim 10, wherein the localizer comprises an optical sensor and the at least one trackable feature comprises at least one optical tracking element, and wherein detecting the actual state of the trackable feature is further defined as: detecting, with the optical sensor, the actual state of the at least one optical tracking element.

12. The method of claim 10, wherein the localizer is a machine vision system and the at least one trackable feature comprises one or more of a predetermined pattern, a shape and a color, and wherein detecting the actual state of the trackable feature is further defined as: detecting, with the machine vision system, the actual state of the one or more of the predetermined pattern, shape and color.

13. The method of claim 11, further comprising: tracking, with the localizer, a state of the surgical device; and obtaining the expected state of the tool by the one or more controllers combining the tracked state of the surgical device and known relationship data between the expected state of the tool and a predetermined reference point on the surgical device.

14. A non-transitory computer-readable medium comprising instructions, which when executed by one or more processors, are configured to: obtain a predetermined state of at least one trackable feature of a trackable protective packaging that is configured to retain a tool, the predetermined state defined relative to an actual state of the tool when the tool is retained by the trackable protective packaging; obtain an expected state of the tool based on an expected condition in which the tool is properly mounted to a surgical device; acquire an actual state of the at least one trackable feature detected by a localizer; determine the actual state of the tool based on the actual state of the at least one trackable feature and the predetermined state of the at least one trackable feature defined relative to the actual state of the tool; compare the actual state and the expected state of the tool; and evaluate whether the tool is properly mounted to the surgical device based on comparison of the actual state and expected state of the tool, wherein the instructions, which when executed by the one or more processors, are configured to generate at least one of a haptic, audible, and visual alert in response to evaluation of whether the tool is properly mounted to the surgical device.

15. The non-transitory computer-readable medium of claim 14, wherein the instructions, which when executed by the one or more processors, acquire the actual state of the at least one trackable feature by further being configured to: acquire the actual state of at least one optical tracking element detected by an optical sensor.

16. The non-transitory computer-readable medium of claim 14, wherein the instructions, which when executed by the one or more processors, acquire the actual state of the at least one trackable feature by further being configured to: acquire the actual state of one or more of a predetermined pattern, shape and color detected by a machine vision system.

17. The non-transitory computer-readable medium of claim 14, wherein the instructions, which when executed by the one or more processors, are configured to: acquire a tracked state of the surgical device from the localizer; obtain known relationship data between the expected state of the tool and a predetermined reference point on the surgical device; and obtain the expected state of the tool by further being configured to combine the tracked state of the surgical device and known relationship data.
